

## SEOGWIPO, Korea, Republic of

WMO: 471890

Lat: 33.2461N

Lon: 126.5653E

Elev: 50

StdP: 100.72

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.4        | 1.7 | -8.4                          | 1.9 | 6.0  | -6.9 | 2.1 | 5.2  | 7.8                   | 8.1  | 7.0 | 8.2  | 2.8                   | 290  | 0.354 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 4.1                    | 31.5            | 26.6 | 30.6 | 26.4 | 29.7 | 26.1 | 27.9                | 29.7 | 27.2 | 29.4 | 26.7 | 29.0 | 3.2                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.5                            | 23.5 | 28.9 | 26.7 | 22.4 | 28.7 | 26.1 | 21.6 | 28.5 | 89.8          | 29.7 | 86.7 | 29.5 | 84.3 | 29.2 | 30.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.5               | 6.4  | 5.5 | -1.2                       | 33.0 | 1.7                | 1.4 | -2.4                                               | 34.0 | -3.4       | 34.9 | -4.3       | 35.7 | -5.6       | 36.7 |    |    |
|                   |      |     | -3.2                       | 28.2 | 1.3                | 1.0 | -4.1                                               | 28.9 | -4.9       | 29.5 | -5.6       | 30.0 | -6.5       | 30.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.1   | 7.2   | 8.2   | 10.9  | 15.0  | 18.9  | 21.9  | 26.0  | 27.5  | 24.4  | 20.0  | 14.7  | 9.5  |
|                                                                                  | DBStd       | 7.28   | 3.05  | 3.18  | 2.94  | 2.37  | 1.93  | 1.59  | 2.28  | 1.72  | 1.93  | 2.38  | 3.19  | 3.24 |
|                                                                                  | HDD10.0     | 238    | 95    | 66    | 24    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 49   |
|                                                                                  | HDD18.3     | 1374   | 344   | 283   | 230   | 101   | 16    | 0     | 0     | 0     | 0     | 11    | 114   | 274  |
|                                                                                  | CDD10.0     | 2825   | 9     | 16    | 53    | 152   | 277   | 358   | 497   | 542   | 431   | 311   | 144   | 33   |
|                                                                                  | CDD18.3     | 918    | 0     | 0     | 0     | 2     | 34    | 108   | 239   | 284   | 181   | 64    | 6     | 0    |
|                                                                                  | CDH23.3     | 6619   | 0     | 0     | 0     | 0     | 42    | 223   | 2092  | 3036  | 1101  | 122   | 2     | 0    |
|                                                                                  | CDH26.7     | 1650   | 0     | 0     | 0     | 0     | 2     | 7     | 519   | 962   | 158   | 3     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.6    | 2.5   | 2.6   | 2.7   | 2.7   | 2.4   | 2.4   | 2.5   | 2.7   | 2.9   | 2.7   | 2.4   | 2.4  |
| Precipitation                                                                    | PrecAvg     | 1789   | 58    | 80    | 113   | 180   | 215   | 280   | 283   | 229   | 161   | 67    | 77    | 45   |
|                                                                                  | PrecMax     | 3244   | 143   | 253   | 218   | 301   | 448   | 896   | 734   | 631   | 611   | 299   | 225   | 164  |
|                                                                                  | PrecMin     | 1146   | 5     | 8     | 22    | 72    | 16    | 63    | 56    | 26    | 24    | 1     | 10    | 3    |
|                                                                                  | PrecStd     | 404    | 40    | 59    | 54    | 57    | 104   | 160   | 157   | 136   | 121   | 64    | 53    | 36   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.0  | 17.4  | 19.1  | 22.4  | 26.2  | 27.7  | 32.7  | 32.7  | 30.9  | 26.8  | 23.3  | 18.8 |
|                                                                                  |             | MCWB   | 14.7  | 15.0  | 14.0  | 16.1  | 16.6  | 20.7  | 26.3  | 26.7  | 25.3  | 20.9  | 19.0  | 15.8 |
|                                                                                  | 2%          | DB     | 14.8  | 15.9  | 17.7  | 20.9  | 24.4  | 26.2  | 31.3  | 31.7  | 29.3  | 25.6  | 21.9  | 17.0 |
|                                                                                  |             | MCWB   | 11.7  | 13.6  | 14.3  | 15.8  | 17.4  | 21.2  | 26.6  | 26.7  | 24.2  | 20.5  | 18.2  | 13.2 |
|                                                                                  | 5%          | DB     | 13.2  | 14.4  | 16.8  | 19.8  | 23.3  | 25.3  | 30.2  | 31.0  | 28.3  | 24.7  | 20.7  | 15.5 |
|                                                                                  |             | MCWB   | 9.7   | 11.3  | 13.5  | 15.2  | 17.5  | 21.6  | 26.5  | 26.6  | 23.8  | 19.8  | 16.9  | 11.7 |
|                                                                                  | 10%         | DB     | 11.7  | 13.1  | 15.6  | 18.9  | 22.2  | 24.6  | 29.2  | 30.3  | 27.5  | 23.7  | 19.6  | 14.3 |
|                                                                                  |             | MCWB   | 8.3   | 9.7   | 12.2  | 15.0  | 17.4  | 21.6  | 26.1  | 26.4  | 23.5  | 19.1  | 15.7  | 10.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.4  | 16.0  | 16.7  | 19.2  | 21.6  | 24.7  | 28.6  | 29.2  | 27.0  | 24.4  | 20.7  | 17.2 |
|                                                                                  |             | MCDB   | 16.6  | 16.9  | 17.5  | 20.3  | 22.2  | 25.3  | 30.0  | 30.3  | 29.1  | 25.4  | 22.2  | 18.0 |
|                                                                                  | 2%          | WB     | 12.2  | 14.2  | 15.6  | 18.1  | 20.4  | 23.9  | 27.7  | 28.2  | 26.1  | 22.8  | 19.2  | 14.2 |
|                                                                                  |             | MCDB   | 14.1  | 15.4  | 16.9  | 19.3  | 21.8  | 24.7  | 29.5  | 29.9  | 27.9  | 24.5  | 21.2  | 15.9 |
|                                                                                  | 5%          | WB     | 10.2  | 12.1  | 14.3  | 17.1  | 19.7  | 23.3  | 27.1  | 27.6  | 25.4  | 21.4  | 17.9  | 12.4 |
|                                                                                  |             | MCDB   | 12.6  | 13.9  | 16.0  | 18.6  | 21.5  | 24.3  | 29.2  | 29.7  | 27.3  | 23.5  | 20.2  | 14.8 |
|                                                                                  | 10%         | WB     | 8.7   | 10.4  | 12.9  | 16.0  | 19.0  | 22.6  | 26.6  | 27.0  | 24.6  | 20.1  | 16.3  | 11.1 |
|                                                                                  |             | MCDB   | 11.3  | 12.6  | 15.4  | 17.8  | 21.1  | 23.8  | 28.6  | 29.4  | 26.6  | 22.9  | 18.9  | 14.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.4   | 5.6   | 5.9   | 5.7   | 5.3   | 4.1   | 3.6   | 4.1   | 4.8   | 5.7   | 5.9   | 5.5  |
|                                                                                  |             | MCDBR  | 6.3   | 6.2   | 6.4   | 6.5   | 6.7   | 5.1   | 4.7   | 4.9   | 5.4   | 6.0   | 6.0   | 6.2  |
|                                                                                  |             | MCWBR  | 5.3   | 5.6   | 5.7   | 5.4   | 5.0   | 3.8   | 3.0   | 3.1   | 3.8   | 4.4   | 5.0   | 5.1  |
|                                                                                  | 5% WB       | MCDBR  | 5.9   | 5.6   | 5.7   | 4.4   | 4.4   | 3.2   | 4.0   | 4.1   | 4.1   | 4.5   | 5.2   | 5.5  |
|                                                                                  |             | MCWBR  | 5.9   | 6.1   | 6.1   | 4.5   | 4.4   | 2.7   | 2.9   | 3.0   | 3.5   | 4.2   | 5.2   | 5.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.462  | 0.501 | 0.551 | 0.561 | 0.567 | 0.595 | 0.577 | 0.534 | 0.494 | 0.466 | 0.459 | 0.454 |      |
|                                                                                  | taud        | 1.962  | 1.900 | 1.795 | 1.828 | 1.867 | 1.891 | 1.985 | 2.117 | 2.168 | 2.109 | 2.048 | 1.985 |      |
|                                                                                  | Ebn at Noon | 712    | 731   | 731   | 749   | 749   | 725   | 736   | 762   | 776   | 759   | 713   | 691   |      |
|                                                                                  | Edh at Noon | 147    | 174   | 208   | 210   | 203   | 198   | 180   | 155   | 142   | 139   | 134   | 137   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.02   | 2.86  | 4.08  | 4.86  | 5.20  | 4.54  | 5.03  | 5.15  | 4.24  | 3.71  | 2.56  | 1.88  |      |
|                                                                                  | RadStd      | 0.24   | 0.36  | 0.32  | 0.41  | 0.58  | 0.57  | 0.71  | 0.51  | 0.40  | 0.37  | 0.29  | 0.17  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page